Chapter 1: Comparison of Batches

Applied Multivariate Statistical Analysis

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Introduction

What Is Multivariate Statistical Analysis?

Setting: We observe n data points in $\mathbb{R}^p$:

$$x_i = (x_{i1}, x_{i2}, \dots, x_{ip}), \quad i = 1, \dots, n,$$

each a realisation of a random vector $X = (X_1, X_2, \dots, X_p)^\top \in \mathbb{R}^p$.

Key questions addressed by descriptive techniques:

- Are some components more spread than others?
- Do sub-groups of observations exist?
- Are there outliers?
- How "normal" is the distribution?
- Are there low-dimensional linear combinations of X with non-normal behaviour?

Perceptual Challenges in High Dimensions

- **2D:** Point clouds are immediately interpretable.
- **3D:** Real-time rotation on a computer screen is feasible.
- **4D:** A “sliding” technique (Härdle & Scott, 1992) can show 3D density contours as the fourth variable varies.
- $\geqslant$ **5D:** A qualitative jump in difficulty occurs; high-dimensional structure must be mapped into lower dimensions (Klinke & Polzehl, 1995).

This chapter introduces:

- Boxplots
- Histograms
- Kernel density estimates
- Scatterplots & draftman plots
- Chernoff–Flury faces
- Andrews’ curves
- Parallel coordinates plots (PCP)

The Swiss Bank Note Dataset

Swiss bank notes (Flury & Riedwyl, 1988): $n = 200$ measurements — 100 **genuine**, 100 **counterfeit**.

Six measurements per note:

$$X_1 = \text{length of bill}$$

$$X_2 = \text{height (left)}$$

$$X_3 = \text{height (right)}$$

$$X_4 = \text{inner frame to lower border}$$

$$X_5 = \text{inner frame to upper border}$$

$$X_6 = \text{diagonal of central picture}$$

Goal: Use these measurements to determine whether a bill is genuine or counterfeit.

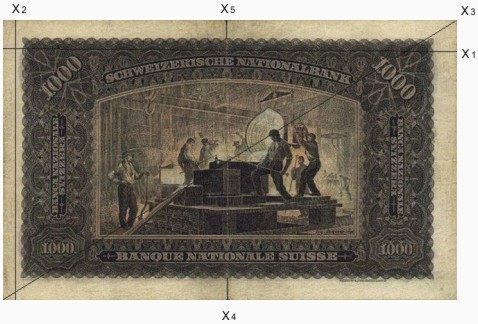


Figure 1: Swiss 1000-franc bank note with measurement locations

Section 1: Boxplots

The Five Number Summary

Example: 15 largest world cities in 2006 (population in 10,000s).

City	Country	Pop. (10k)	Order stat.
Tokyo	Japan	3,420	$x_{(15)}$
Mexico City	Mexico	2,280	$x_{(14)}$
Seoul	S. Korea	2,230	$x_{(13)}$
New York	USA	2,190	$x_{(12)}$
Sao Paulo	Brazil	2,020	$x_{(11)}$
Bombay	India	1,985	$x_{(10)}$
Delhi	India	1,970	$x_{(9)}$
Shanghai	China	1,815	$x_{(8)} \leftarrow M$
$\vdots$		$\vdots$	$\vdots$
Karachi	Pakistan	1,430	$x_{(1)}$

Median ($n = 15$ odd): $M = x_{(8)} = 1{,}815$.

Quartiles and F-Spread

Depth of the fourth: $\dfrac{[\text{depth of median}]+1}{2} = \dfrac{8+1}{2} = 4.5$

So the **fourths** (quartiles) are:

$$F_L = \tfrac{1}{2}\{x_{(4)} + x_{(5)}\} = 1{,}610, \qquad F_U = \tfrac{1}{2}\{x_{(11)} + x_{(12)}\} = 2{,}105.$$

F-**spread** and **outside bars** (outlier thresholds):

$$d_F = F_U - F_L = 495$$
$$F_L - 1.5\, d_F = 1{,}610 - 742.5 = 867.5$$
$$F_U + 1.5\, d_F = 2{,}105 + 742.5 = 2{,}847.5$$

$\Rightarrow$ **Tokyo** (3,420) is beyond the upper outside bar $\Rightarrow$ **outlier**.

Five Number Summary: $\{1{,}430;\ 1{,}610;\ 1{,}815;\ 2{,}105;\ 3{,}420\}$. Mean: $\bar{x} = 1{,}939.7$.

Constructing the Boxplot

Algorithm:

1. Draw a **box** with borders at F_L and F_U (contains 50% of the data).
2. Draw the **median** as a solid line $|$ and the **mean** as a dotted line $\vdots$.
3. Draw **whiskers** from each box end to the most remote point that is *not* an outlier.
4. Mark outliers as $\star$ (outside $F_{UL} \pm 1.5\, d_F$) or $\bullet$ (outside $F_{UL} \pm 3\, d_F$).

Constructing the Boxplot

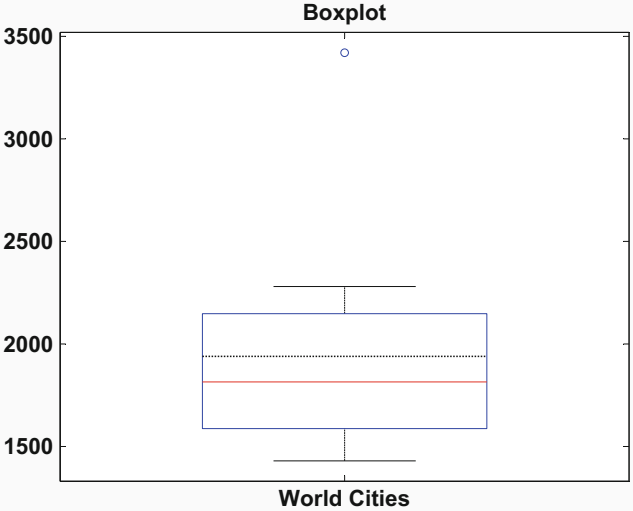
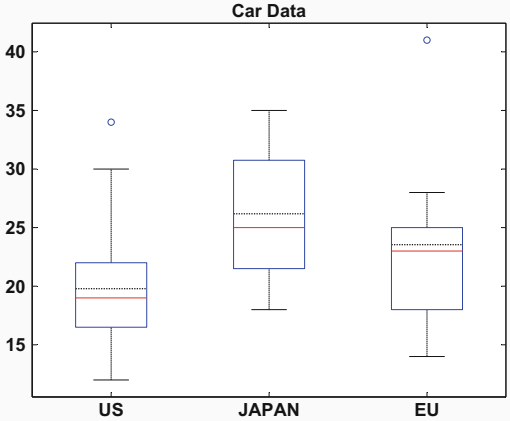


Figure 2: Boxplot for world cities. One outlier (Tokyo) is visible; the data are right-skewed.

Comparing Car Fuel Economies



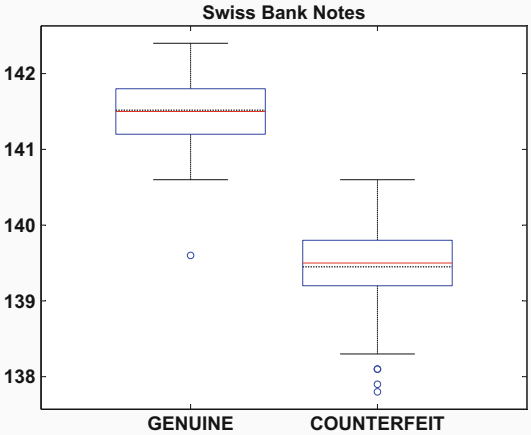
Five Number Summaries:

Group	Min	F_L	M	F_U	Max
US	12	16.8	18.8	22	30
Japan	18	22	25	30.5	35
EU	14	19	23	25	28

Conclusions:

- Japanese cars are most fuel-efficient.
- One EU outlier: VW Rabbit Golf Diesel.
- US box lies entirely below Japanese box.
- Worst Japanese car outperforms $\approx 50\%$ of US cars.

Swiss Bank Notes: X_6 (Diagonal)



Genuine notes ($n = 100$):

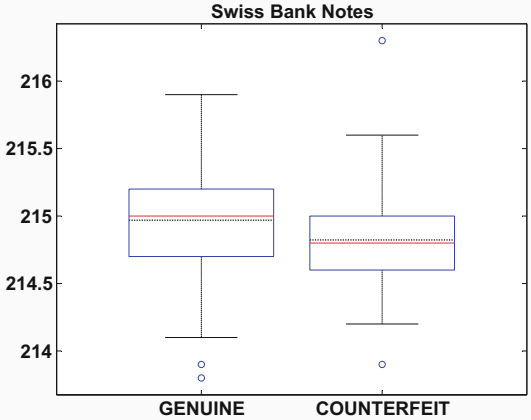
#	100	Genuine	
M	50.5	141.5	
F	25.75	141.25	141.8
	1	140.65	142.4

Counterfeit notes ($n = 100$):

#	100	Counterfeit	
M	50.5	139.5	
F	25.75	139.2	139.8
	1	138.3	140.65

$\Rightarrow$ Genuine diagonals are **systematically larger**, but with some overlap.

Swiss Bank Notes: X_1 (Length)



- Length (X_1) shows **less clear separation** than diagonal (X_6).
- Distributions largely overlap.
- A few outliers in each group.
- Boxplots alone do **not reveal multimodality**.

Boxplot Summary

Summary: Boxplots	
↪	Median and mean are measures of **location**.
↪	Position of median/mean within the box measures **skewness**.
↪	Box length and whisker length measure **spread** and **tail length**.
↪	Outliers: $\star$ outside $F_{UL} \pm 1.5d_F$; $\bullet$ outside $F_{UL} \pm 3d_F$.
↪	Boxplots do **not** reveal multimodality or clusters.
↪	Side-by-side boxplots compare entire **distributions**.

Section 2: Histograms

Definition of the Histogram

Bin of length h with origin x_0:

$$B_j(x_0, h) = [x_0 + (j-1)h,\ x_0 + jh), \quad j \in \mathbb{Z}.$$

Histogram estimator for an i.i.d. sample $\{x_i\}_{i=1}^n$:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{j \in \mathbb{Z}} \sum_{i=1}^{n} \mathbf{1}\{x_i \in B_j(x_0, h)\}\, \mathbf{1}\{x \in B_j(x_0, h)\}.$$

Key parameter: binwidth h.

- h too **large** $\Rightarrow$ over-smoothed, unstructured histogram.
- h too **small** $\Rightarrow$ noisy histogram with many spurious peaks.

Optimal binwidth (rule of thumb): $h_{\text{opt}} = \left(\frac{24\sqrt{\pi}}{n}\right)^{1/3}$.

Effect of Binwidth h

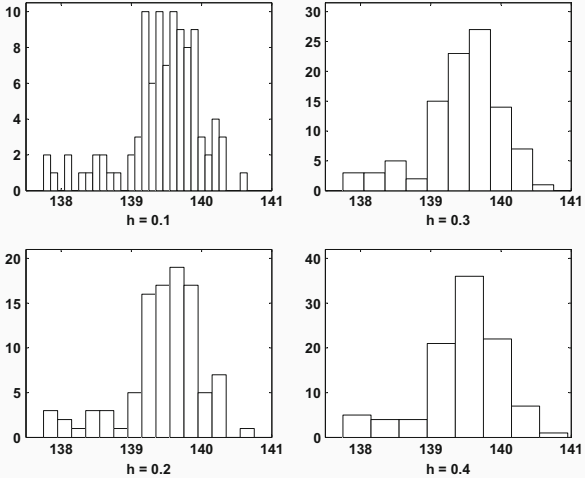


Figure 3: Diagonal of counterfeit bank notes. Histograms with $x_0 = 137.8$ and $h = 0.1$ (upper left), $h = 0.2$ (lower left), $h = 0.3$ (upper right), $h = 0.4$ (lower right).

Effect of Bin Origin x_0

Problem: Even with fixed h, different origins x_0 yield different histograms!

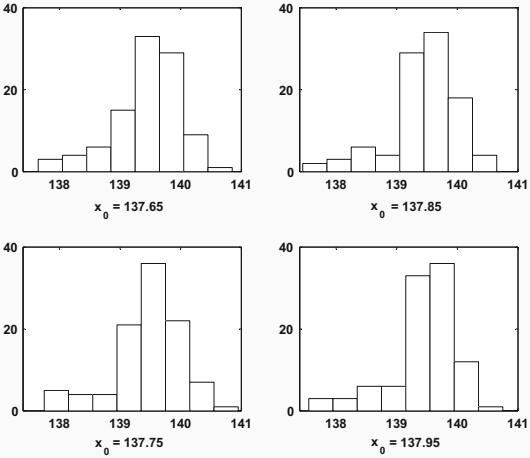


Figure 4: Diagonal of counterfeit notes. Fixed $h = 0.4$; origins $x_0 \in \{137.65, 137.75, 137.85, 137.95\}$.

Averaged Shifted Histograms (ASH)

Averaging over multiple histogram origins removes the dependence on x_0 and reveals the **bimodal** structure of the diagonal variable.

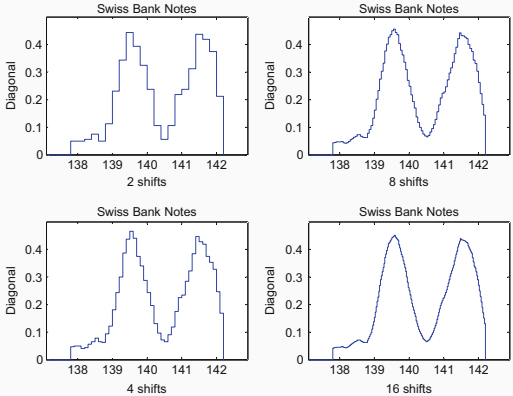


Figure 5: ASH based on all Swiss bank notes: 2 shifts (upper left), 4 shifts (lower left), 8 shifts (upper right), 16 shifts (lower right).

Histogram Summary

Summary: Histograms	
$\hookrightarrow$	Histograms detect **modes**; modes correspond to strong peaks.
$\hookrightarrow$	Same h, different $x_0 \Rightarrow$ different histograms.
$\hookrightarrow$	Changing x_0 can drastically alter appearance.
$\hookrightarrow$	h too large $\Rightarrow$ flat, unstructured; h too small $\Rightarrow$ unstable.
$\hookrightarrow$	Optimal $h_{\text{opt}} = (24\sqrt{\pi}/n)^{1/3}$ (under normality).
$\hookrightarrow$	Averaged shifted histograms are kernel density estimates.

Section 3: Kernel Densities

From Histogram to Kernel Density

Four critiques of histograms:

1. Binwidth h is hard to choose optimally.
2. Origin x_0 also influences the shape.
3. Information is lost (observations replaced by bin centre).
4. The underlying density is smooth, but the histogram is not.

Key insight: The histogram at a bin centre can be written as:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{x - x_i}{h}\Big), \quad K(u) = \mathbf{1}\big(|u| \leq \tfrac{1}{2}\big).$$

Replacing the uniform kernel K with a smooth function resolves difficulties (2)–(4).

Common Kernel Functions

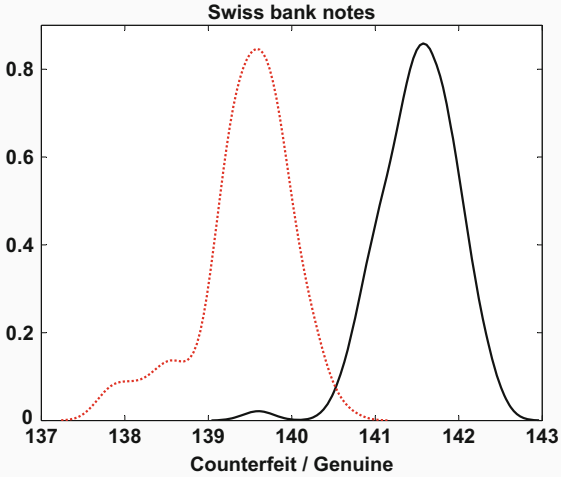
$K(\bullet)$	Name
$K(u) = \frac{1}{2}\,\mathbf{1}(\lvert u\rvert \leqslant 1)$	Uniform
$K(u) = (1-\lvert u\rvert)\,\mathbf{1}(\lvert u\rvert \leqslant 1)$	Triangle
$K(u) = \frac{3}{4}(1-u^2)\,\mathbf{1}(\lvert u\rvert \leqslant 1)$	Epanechnikov
$K(u) = \frac{15}{16}(1-u^2)^2\,\mathbf{1}(\lvert u\rvert \leqslant 1)$	Quartic (Biweight)
$K(u) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u^2}{2}\right)$	Gaussian φ

Bandwidth selection (Silverman's rule of thumb for Gaussian kernel):

$$h_G = 1.06\,\hat{\sigma}\,n^{-1/5}, \qquad \hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2}.$$

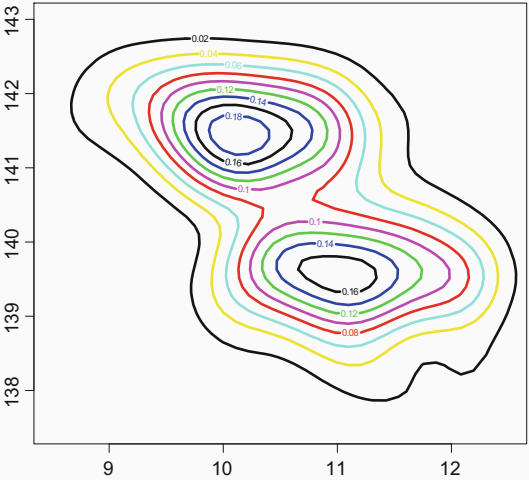
For the quartic kernel: $h_Q = 2.62 \cdot h_G$.

KDE of Bank Note Diagonals

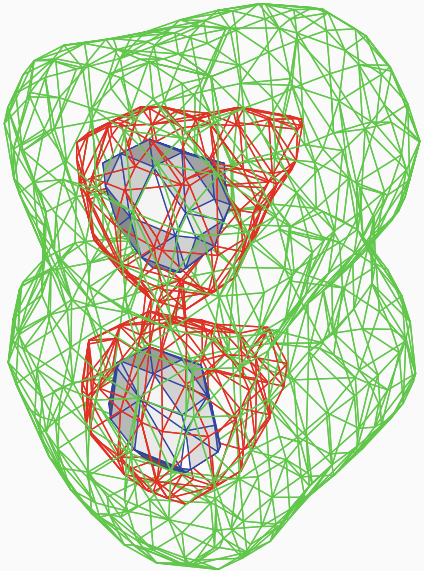


- **Black (solid):** Genuine notes ($X_6 \approx 141.5$).
- **Red (dotted):** Counterfeit notes ($X_6 \approx 139.5$).
- Clear separation, but **overlap** near ≈ 140.4.
- Cannot perfectly discriminate using X_6 alone.
- Need additional variables: X_4, X_5, ...

Bivariate and Trivariate Density Estimates



2D: X_5 vs X_6

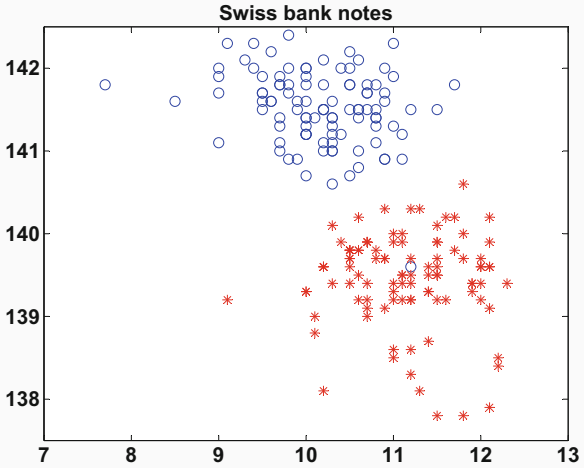


KDE Summary

Summary: Kernel Density Estimates	
$\hookrightarrow$	KDEs estimate densities by centring a smooth kernel on each observation.
$\hookrightarrow$	Bandwidth h controls smoothness of $\widehat{f}$.
$\hookrightarrow$	KDEs are smooth and can represent distributions up to 3D.
$\hookrightarrow$	Rule of thumb: $h_G = 1.06\hat{\sigma}n^{-1/5}$ (Gaussian kernel only).
$\hookrightarrow$	Good for detecting modes, location, skewness, tails, asymmetry.

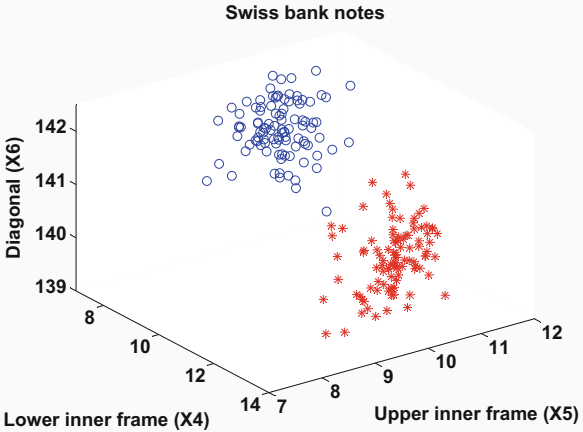
Section 4: Scatterplots

2D Scatterplot: X_5 vs X_6



- $\circ$ = genuine notes; $*$ = counterfeit notes.
- **Downward-sloping scatter**: larger X_5 $\Rightarrow$ smaller X_6.
- The **70th observation** (thick circle) is a genuine note embedded in the counterfeit cloud.
- A horizontal separation line would misclassify it.
- Better separation requires **more variables**.

3D Scatterplot: X_4, X_5, X_6



- Adding X_4 (lower inner frame distance) gives **better cluster separation**.
- The separation surface is now a **plane**, not a line:

$$a_1 x_1 + \cdots + a_6 x_6 = \text{const.}$$

- The optimal weights $(a_1, \ldots, a_6)$ are found by **discriminant analysis** (Chapter 14).

Draftman's Plot (Scatterplot Matrix)

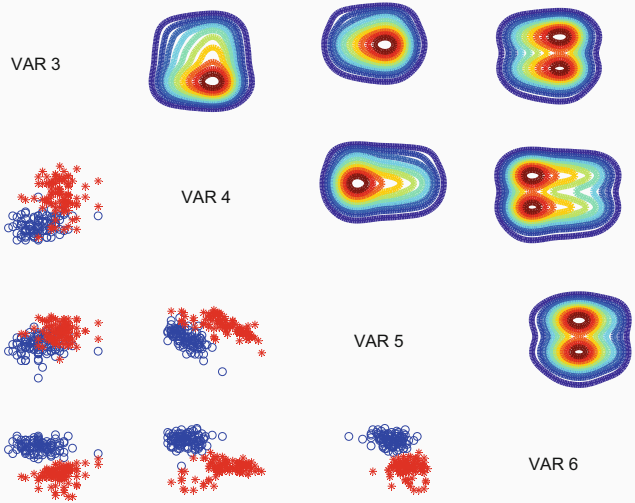


Figure 6: Draftman's plot: X_3, X_4, X_5, X_6 of the bank data. Upper right half shows density contour plots.

Scatterplot Summary

Summary: Scatterplots	
↪	2D/3D scatterplots identify separated points, outliers, sub-clusters.
↪	Slope of scatter indicates positive/negative dependence.
↪	Draftman plot matrices detect structures conditioned on other variables.
↪	Brushing through a point cloud reveals conditional dependence.

Section 5: Chernoff–Flury Faces

The Idea of Faces (Chernoff 1973, Flury & Riedwyl 1988)

- **Principle:** Map each p-dimensional observation onto characteristics of a human face.
- Each variable is rescaled to $[0, 1]$; the two extremes correspond to extreme face element positions.
- Flury–Riedwyl design: **36 face elements** (18 for left side, 18 for right side).
- **Bank data assignment** ($p = 6$):

Variable	Face Element
X_1	Eye sizes (1, 19)
X_2	Pupil sizes (2, 20)
X_3	Eye slants (4, 22)
X_4	Upper hair lines (11, 29)
X_5	Lower hair lines (12, 30)
X_6	Face lines + darkness of hair (13, 14, 31, 32)

Faces for Observations 91–110

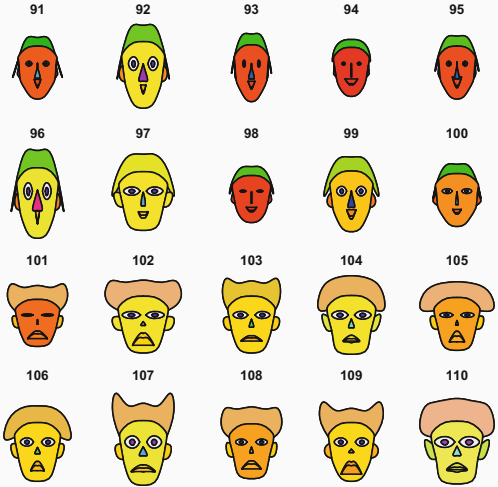
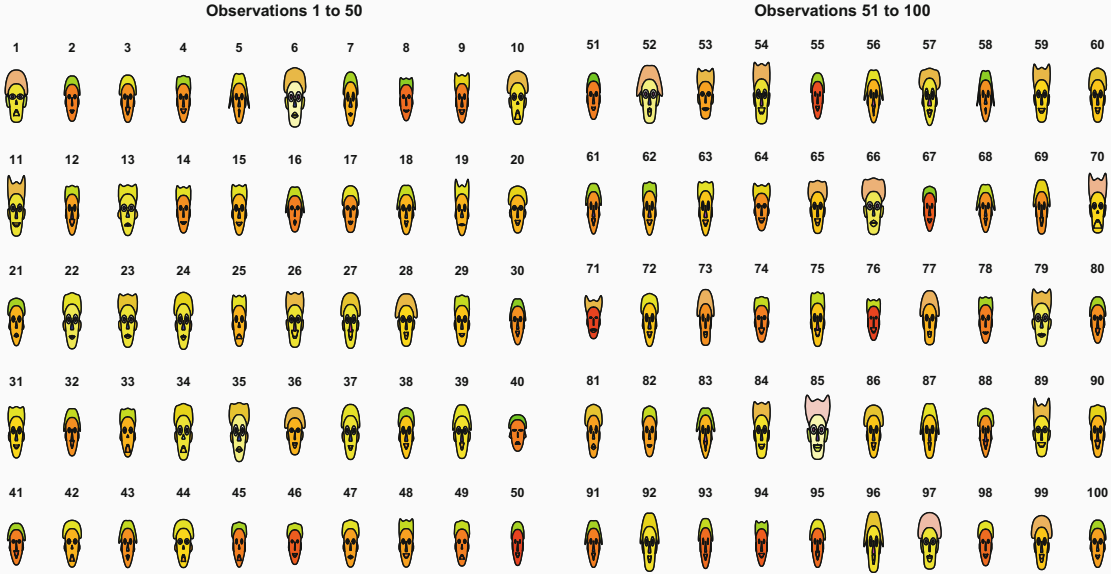
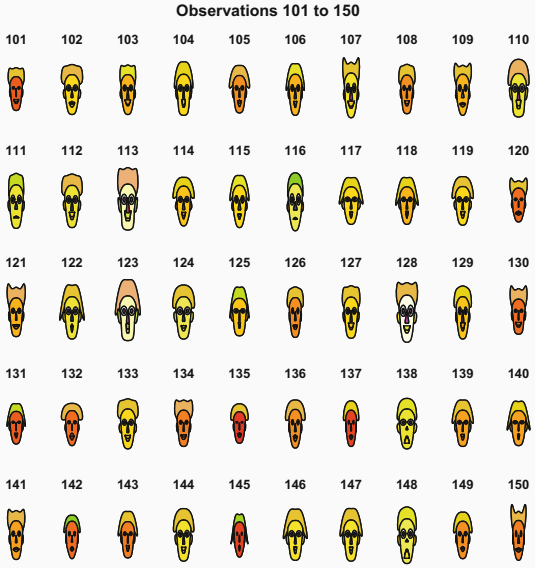


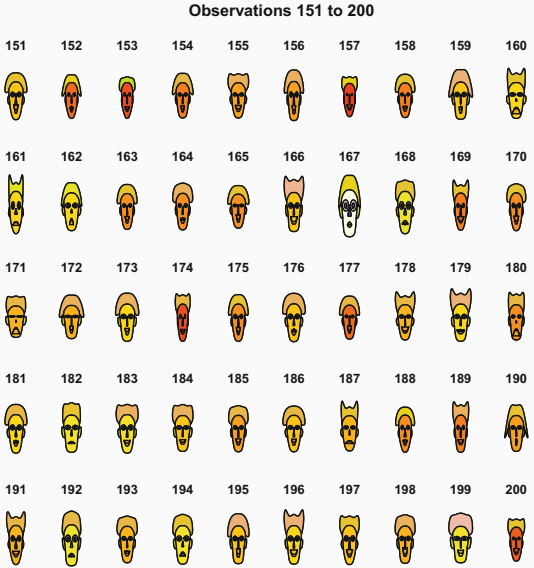
Figure 7: Chernoff–Flury faces for observations 91–110 of the bank data. Observations 91–100: **counterfeit** (upper); 101–110: **genuine** (lower).

Faces for All Genuine Notes (Obs. 1–100)



Faces for All Counterfeit Notes (Obs. 101–200)





Faces Summary

Summary: Chernoff–Flury Faces	
↪	Sub-groups are characterised by **similar-looking faces**.
↪	Outliers appear as **extreme faces** (e.g. dark hair, big smile).
↪	If one element of X is unusual, the corresponding face element changes dramatically.
↪	Variable assignment to face elements matters for interpretability.

Section 6: Andrews' Curves

Definition

Idea (Andrews, 1972): Map each p-dimensional observation X_i to a Fourier curve:

$$f_i(t) = \begin{cases} \dfrac{X_{i,1}}{\sqrt{2}} + X_{i,2}\sin t + X_{i,3}\cos t + \cdots + X_{i,p}\sin\left(\frac{p}{2}t\right) & p \text{ even} \\ \dfrac{X_{i,1}}{\sqrt{2}} + X_{i,2}\sin t + X_{i,3}\cos t + \cdots + X_{i,p}\cos\left(\frac{p-1}{2}t\right) & p \text{ odd} \end{cases}$$

where $t \in [-\pi, \pi]$.

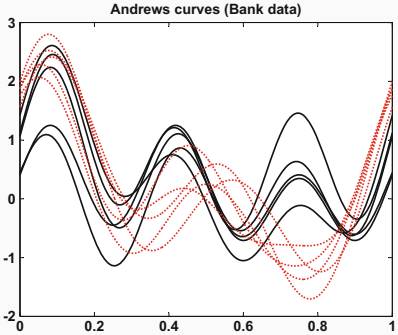
Example ($p = 3$): $X_1 = (0, 0, 1)$, $X_2 = (1, 0, 0)$, $X_3 = (0, 1, 0)$ give

$$f_1(t) = \cos t, \quad f_2(t) = \tfrac{1}{\sqrt{2}}, \quad f_3(t) = \sin t. \quad \text{(Distinct!)}$$

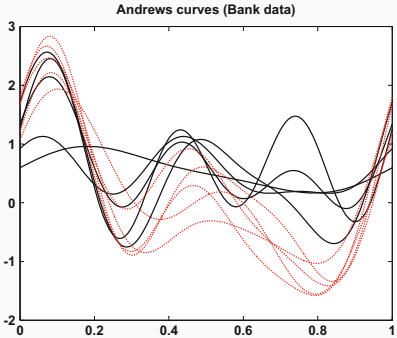
Bank note example (obs. 96, order $1, 2, \ldots, 6$):

$$f_{96}(t) = \tfrac{215.6}{\sqrt{2}} + 129.9\sin t + 129.9\cos t + 9.0\sin 2t + 9.5\cos 2t + 141.7\sin 3t.$$

Andrews' Curves: Variable Order Matters



Order: $1, 2, 3, 4, 5, 6$



Order: $6, 5, 4, 3, 2, 1$

- **Black:** genuine (obs. 96–100); **Red dotted:** counterfeit (obs. 101–105).
- Original order: differences appear in **high frequency** (hard to see).
- Reversed order: differences shift to **intercept** level.
- **Andrews' recommendation:** use order suggested by **Principal Component Analysis**.

Andrews' Curves Summary

Summary: Andrews' Curves	
$\hookrightarrow$	Outliers appear as single curves **deviating from the bundle**.
$\hookrightarrow$	Sub-groups are characterised by **sets of similar curves**.
$\hookrightarrow$	The **order of variables** strongly affects interpretation.
$\hookrightarrow$	Optimal ordering is provided by **PCA** (Chapter 11).
$\hookrightarrow$	For > 20 observations: bad signal-to-ink ratio (Tufte, 1983).

Section 7: Parallel Coordinates Plots

PCP: Concept and Construction

Parallel Coordinates Plot (PCP) (Inselberg, 1985):

- Draw p parallel vertical axes (one per variable).
- Scale all variables to $[\min, \max] = [0, 1]$.
- Represent each observation x_i as a **polyline** connecting its scaled values on the p axes.

Advantages over Cartesian plots:

- Works for $p > 4$ (no dimensionality barrier).
- Positive dependence: lines approximately **parallel**.
- Negative dependence: lines **intersect** near the midpoint.
- Discrete variables and sub-groups are **visible**.

PCP for Bank Note Observations 96–105

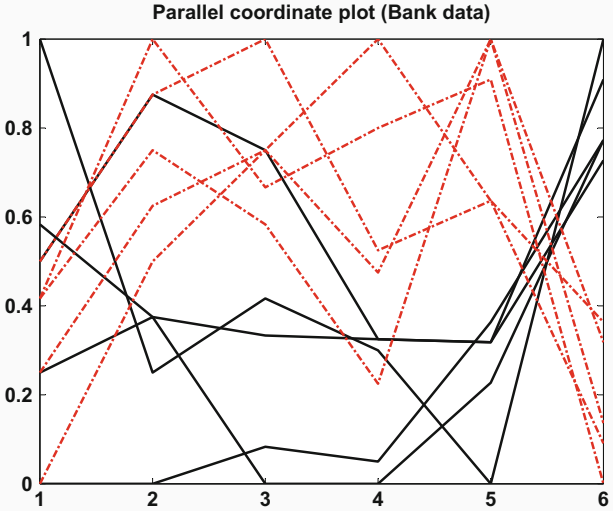
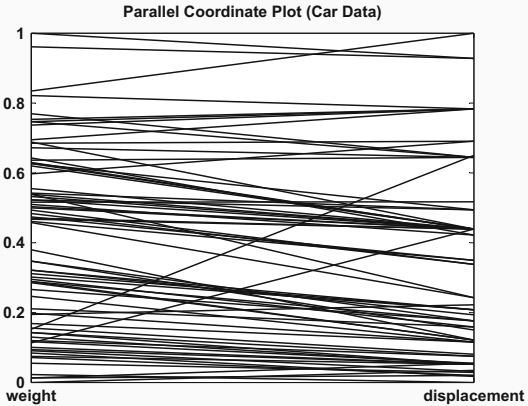
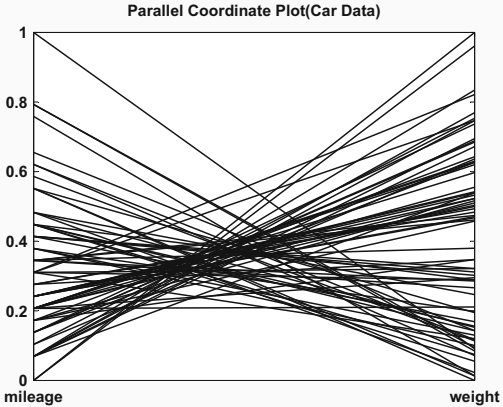


Figure 8: PCP of observations 96–105 (genuine: black solid; counterfeit: red dash-dot).

Detecting Linear Dependence



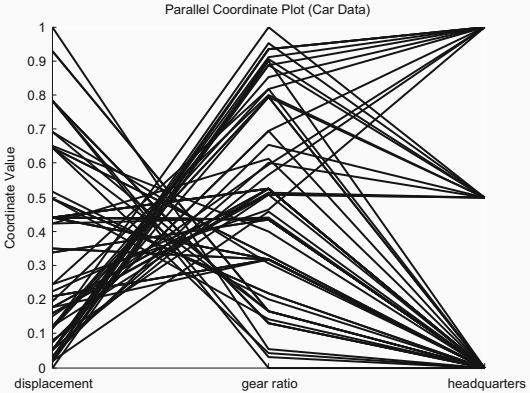
Weight vs Displacement ($\rho = 0.9$)



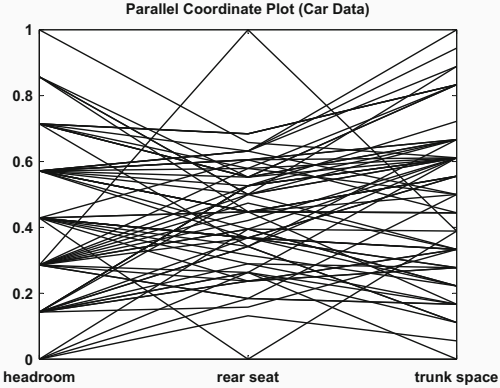
Mileage vs Weight ($\rho = -0.82$)

- **Near-parallel lines** $\Rightarrow$ strong **positive** correlation.
- **Lines crossing near midpoint** $\Rightarrow$ strong **negative** correlation.

PCP for Sub-Group Detection and Outliers



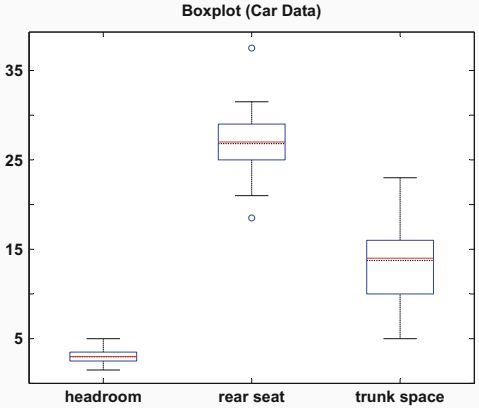
Displacement, Gear Ratio, Headquarters

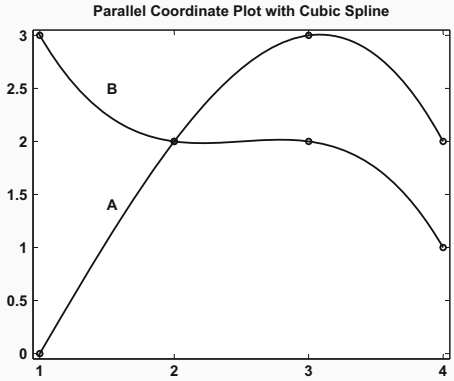


Headroom, Rear Seat, Trunk Space

- **Left:** Convergence to 3 discrete headquarters values (USA / Japan / EU).
- **Right:** Two **outliers** clearly visible — confirmed by the boxplot below.

Boxplot Confirmation and Cubic Spline PCP





Cubic spline PCP

- **Crossing ambiguity** in standard PCP (obs. A and B sharing a value at $j = 2$): unresolvable with straight lines.
- **Cubic spline** interpolation resolves which curve belongs to which observation.

Full Car Data PCP with Colouring

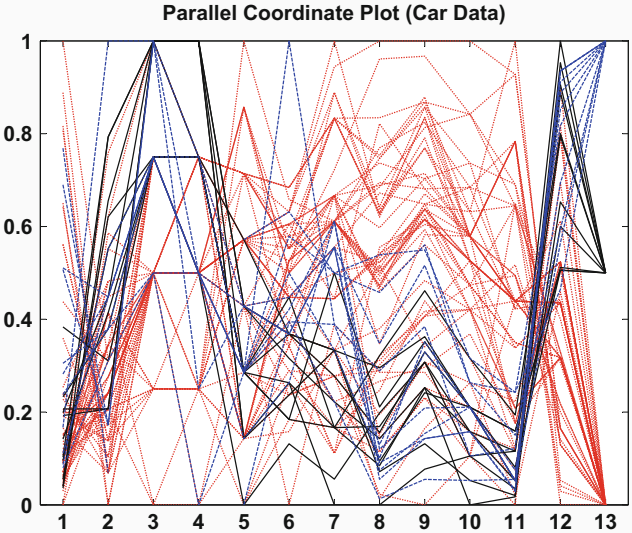


Figure 9: PCP for all 13 car data variables, coloured by headquarters: red = EU, blue = Japan, black = US.

PCP Summary

Summary: Parallel Coordinates Plots	
↪	PCP overcomes the Cartesian limitation for dimensions > 4.
↪	Outliers appear as **outlying polygon curves**.
↪	Near-parallel lines indicate **positive** dependence; crossing lines indicate **negative** dependence.
↪	Variable ordering matters; sub-groups may be revealed by selective **colouring**.
↪	Cubic spline interpolation resolves crossing ambiguity.

Section 8: Hexagon Plots

Motivation and Algorithm

Problem: For large n, ordinary scatterplots suffer from **overplotting** (many points on top of each other).

Hexagon binning (Lewin-Koh, 2006):

1. Tessellate the (x, y)-plane with a regular grid of hexagons.
2. Count the number of points n_k in each hexagon k.
3. Colour or scale hexagons in proportion to n_k.

Advantages of hexagons over squares:

- **Symmetry of nearest neighbours** (equidistant centres).
- Maximum number of sides for a regular plane tessellation.
- Less visually biased for displaying densities.
- Extremely fast — works for $n \geqslant 10^6$.

Hexagon Plots: Age vs Income and Age vs Computer Time

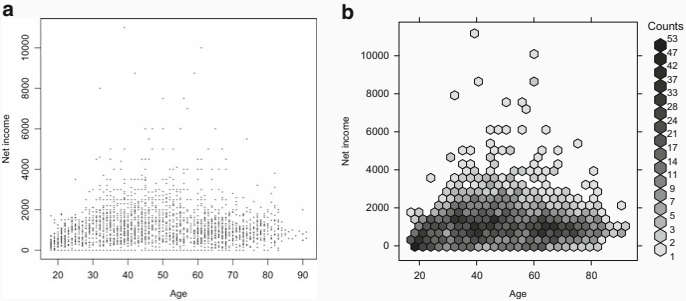


Figure 10: X_1 (Age) vs X_2 (Net income)

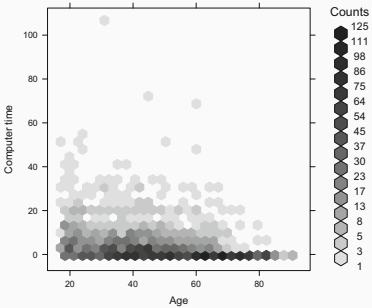


Figure 11: X_1 (Age) vs X_5 (Computer time)

ALLBUS 2006 survey data ($n = 2{,}946$).

Most respondents earn $< 2{,}000$ EUR/month; $\approx 40\%$ of all age groups use a computer < 1 h/week.

Hexagon Plot: Income vs Flat Size

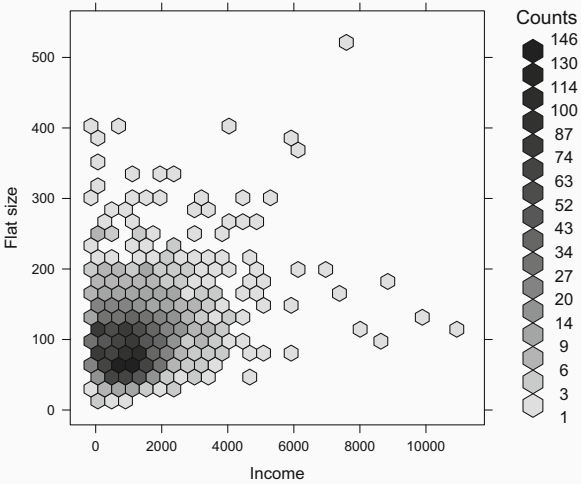


Figure 12: Hexagon plot: X_2 (Net income) vs X_7 (Flat size). Clear positive trend: Higher net income $\Rightarrow$ larger flat. A few high-income respondents live in small flats.

Hexagon Plot Summary

Summary: Hexagon Plots	
↪	Hexagon binning is a **bivariate histogram** for large data.
↪	Variance and bias trade off in opposite directions with bin width.
↪	Hexagons have the property of **symmetry of nearest neighbours**.
↪	Hexagons are visually **less biased** for density display than squares.

Section 9: Boston Housing

Dataset Overview

Boston Housing (Harrison & Rubinfeld, 1978):

- **Goal:** Does "clean air" influence house prices?
- $n = 506$ census tracts in the Boston metropolitan area.
- 14 variables $X_1, \dots, X_{14}$ (see Chapter 22 for full description).
- X_{14}: median value of owner-occupied homes (target).

Key variables: X_1 crime rate, X_3 non-retail business, X_5 nitric oxide concentration, X_6 rooms/dwelling, X_{11} pupil/teacher ratio, X_{13} lower-status population, X_{14} median home value.

PCP for Boston Housing

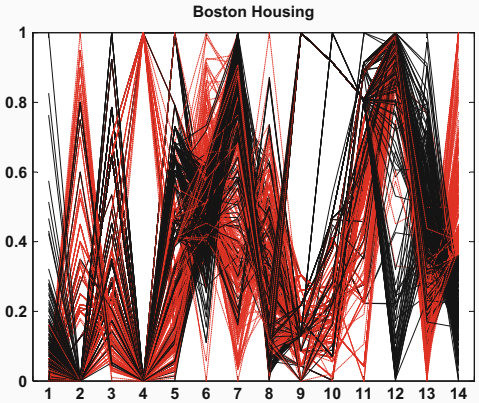
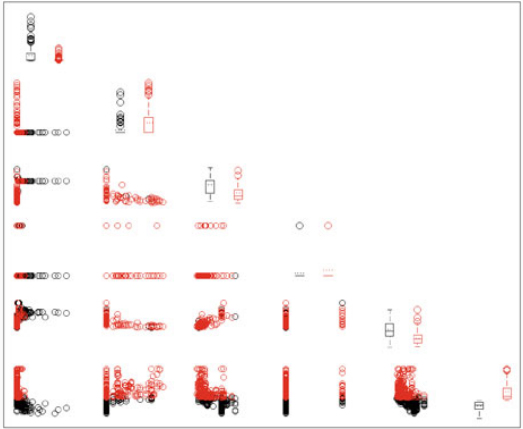


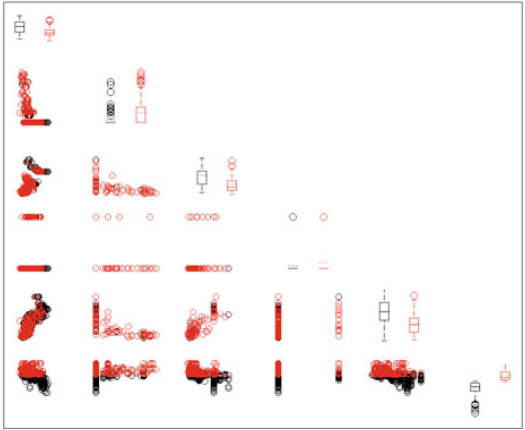
Figure 13: PCP of Boston housing data. Observations with $X_{14} >$ median(X_{14}) are coloured red.

- Strong **negative** dependence: X_{13} and X_{14}.
- Red lines dominate the **upper** part of X_{12}, **lower** part of X_{11}.
- X_1 and X_9: values **concentrated near 0** $\Rightarrow$ transformations needed.

Scatterplot Matrices (Original and Transformed)



$X_1, \ldots, X_5, X_{14}$



$\tilde{X}_1, \ldots, \tilde{X}_5, \tilde{X}_{14}$

- **Right:** Transformed data show more linear relationships.
- Black = cheaper districts; red = expensive districts.

Proposed Transformations

To reduce asymmetry in the marginal distributions:

$$\tilde{X}_1 = \log X_1, \qquad \tilde{X}_2 = X_2/10, \qquad \tilde{X}_3 = \log X_3,$$
$$\tilde{X}_5 = \log X_5, \qquad \tilde{X}_6 = \log X_6, \qquad \tilde{X}_7 = X_7^{2.5}/10000,$$
$$\tilde{X}_8 = \log X_8, \qquad \tilde{X}_9 = \log X_9, \qquad \tilde{X}_{10} = \log X_{10},$$
$$\tilde{X}_{11} = e^{0.4X_{11}}/1000, \qquad \tilde{X}_{12} = X_{12}/100, \qquad \tilde{X}_{13} = \sqrt{X_{13}},$$
$$\tilde{X}_{14} = \log X_{14}.$$

Note: X_4 is binary — **no transformation**.

Boxplots Before and After Transformation

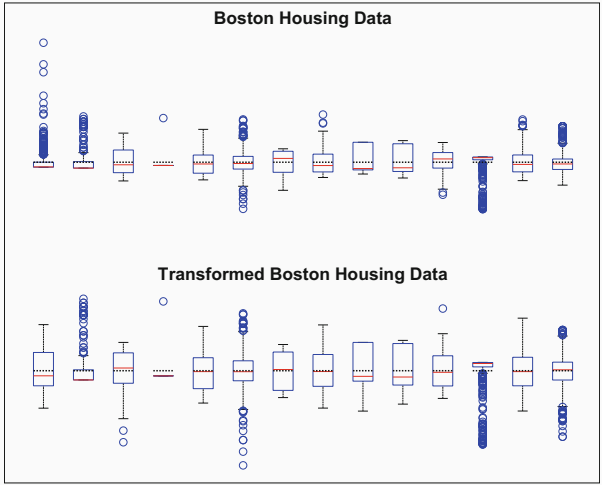


Figure 14: Boxplots for all variables before (top) and after (bottom) transformation: Transformed variables are more symmetric and have fewer outliers.

Key Variable Relationships

Variable	Relation to X_{14}	Remark
X_1 (crime)	Negative	Two sub-groups visible in $\log X_1$
X_3 (non-retail)	Negative	Bimodal KDE
X_5 (NOx)	Negative	May proxy noise pollution
X_6 (rooms)	**Positive**	Strongest linear signal
X_{11} (pupil/teacher)	Negative	Fewer teachers $\Rightarrow$ lower prices
X_{13} (lower status)	**Strongest negative**	Nearly linear on log scale